

Ximing Tao

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Summary

AI Engineer with 5 years of experience building applied AI systems across workflow automation, document intelligence, RAG, structured extraction, and LLM evaluation. Strong at turning ambiguous business processes into structured rules, validation workflows, and deployable automation pipelines. Hands-on across Python, PostgreSQL, LangChain, LangSmith, ChromaDB, Power Automate, AWS, Docker, OCR/layout recognition, and full-stack AI application development.

Skill Highlights

Languages

Python, TypeScript/JavaScript, SQL

AI / LLM

RAG, LangChain, LangSmith, ChromaDB, Prompt Engineering, Context Engineering, Harness Engineering, LLM Evaluation

ML / Document AI

OCR, PaddleOCR, Faster R-CNN, OpenCV, AWS Textract

Full-stack & Automation

PostgreSQL, REST APIs, Django, FastAPI, Node.js, React, Next.js, Power Automate, n8n

Cloud & DevOps

AWS (EC2, S3, Lambda, SageMaker, Bedrock), Azure (Azure AI Document Intelligence, Azure OpenAI Service, Azure Functions, Azure Container Apps), Docker, Git, CI/CD

Work Experience

AI & Automation Engineer

Oct 2025 - May 2026

Xperience

Melbourne

Project 1: Sales Order Automation

- Architected and delivered a cloud-hosted sales order automation platform for Isalbi, using Power Automate triggers, Azure Container Apps, Azure Service Bus, SharePoint, PostgreSQL, and SFTP handoff — saving ~2 hours per Sales Support user per day and achieving ~70% fully automatic processing for eligible orders in production.
- Built the deterministic worker pipeline for Microsoft Graph email ingestion, PDF extraction with Azure Document Intelligence, script-based parsing for eight fixed Excel templates (.xlsx/.xlsm), standard order JSON normalisation, and AX-compatible CSV export.
- Isolated LLM capabilities into a dedicated ai-service for email order classification, PDF JSON extraction, unknown Excel edge-case classification, master-data fuzzy ranking, exception summarisation, and clarification email drafting, while keeping Zendesk creation, CSV generation, SFTP upload, and approval-gated email dispatch deterministic.
- Implemented strict validation gates against versioned customer address, delivery instruction, and item master data synced from SFTP into PostgreSQL, covering PO integrity, quantity, MOQ, price, and delivery-date lead time; routed mismatches to Zendesk review and SS-approved clarification workflows.

AI Engineer (Freelance)

Mar 2025 – Sep 2025; May 2026 – Present

Project 1: AI-Powered Document Intelligence Platform MVP

- Faced with high inference costs and mixed document complexity in a document-processing MVP, extended a Next.js SaaS starter into a cost-aware document intelligence platform and reduced AI processing cost by over 80%.
- Designed a local-first routing pipeline with document classification, confidence-based escalation, prompt routing, structured extraction, validation logic, and error-handling flows across mixed document types.

- Evaluated routing decisions against representative document samples to compare cost, extraction quality, and failure cases, escalating complex or low-confidence cases to higher-capability models.

Project 2: Chatbot & Scheduling System for Child Coding School

- Built an AI-assisted enquiry and scheduling system for a child coding school where parent questions, class availability, and booking changes were previously handled manually.
- Implemented RAG-based enquiry handling and SQL-backed workflow logic to answer parent questions, match students to suitable classes, and manage class availability.
- Designed PostgreSQL constraint-based scheduling rules to prevent overbooking, support 50+ active weekly users, and reduce repetitive admin workload by approximately 60% through automated availability checks, class matching, and booking conflict handling.

Project 3: Automation Medical Practice Assistant

- To reduce missed patient communication and manual note capture, built a context-aware automation assistant integrating calendar, messaging, and structured AI outputs, lowering missed patient responses from 5–8 per day to under 2.

Machine Learning Engineer

Virtual Mike

Apr 2024 - Mar 2025

Australia

Project 1: Real Estate Law Chatbot and LLM Evaluation Tool

- Built a controlled legal Q&A system that mapped natural-language real estate law enquiries to approved, versioned PostgreSQL answer records, ensuring final responses were returned only from verified legal content rather than freely generated by an LLM.
- Designed the query-understanding layer to extract structured retrieval signals such as legal topic, user intent, jurisdiction, timing, and uncertainty level before matching against the approved answer knowledge base.
- Implemented semantic question matching with ChromaDB and metadata-controlled answer retrieval from PostgreSQL, with human fallback for ambiguous, low-confidence, or unsupported enquiries.
- Implemented a rule-based multi-LLM routing strategy with LangChain to balance cost, latency, and query complexity across different legal query categories.
- Built a LangSmith-based evaluation dashboard to monitor cost, latency, token usage, retrieval behaviour, confidence/fallback patterns, and expert review outcomes, supporting data-driven model and prompt selection.
- Reduced repetitive legal review effort by approximately 70% by routing common enquiries to verified answer retrieval and escalating ambiguous or high-risk cases to human review.

Machine Learning Engineer

Auxilis

Apr 2022 - Feb 2024

Australia

Project 1: Document Intelligence Platform

- To reduce reliance on AWS Textract for complex business documents, developed a unified PaddleOCR-based document intelligence system for end-to-end layout analysis across tables, paragraphs, and titles.
- Built a 15K-sample training dataset through a semi-automated labeling pipeline using AWS Textract and Label Studio, improving overall layout parsing accuracy from a 65% baseline to 85%.
- Implemented rule-based post-processors for structural correction, pushing final extraction precision to 92% and reducing reliance on expensive cloud APIs by 60%.
- Developed a specialized Faster R-CNN table detector for challenging table structures, reaching 98% precision and contributing its training data and learnings to the final unified layout model.

Project 2: Handwritten Bengali Digit Recognition

- Designed an end-to-end OCR pipeline for 8 fixed-structure document types using template-based region detection with OpenCV.
- Trained and deployed a Fast R-CNN model (TensorFlow) specifically to recognize and resolve overlapping handwritten digits, a key challenge for the client.
- Achieved real-time performance by leveraging the lightweight architecture of Fast R-CNN and implementing an optimized batch-processing inference strategy, meeting all client benchmarks for speed and accuracy under noisy scan conditions.

Machine Learning Engineer
Wuhan Purvar Bigdata Technology Co., Ltd.

Jun 2020- Sep 2021
China

Project 1: Essay Auto Scoring System

- Worked on an English essay auto-scoring system where long-form student writing needed to be scored consistently against human-marked samples.
- Fine-tuned a BERT-based Transformer model to generate essay-level representations for automated scoring and tested regression/ranking-loss approaches to improve score consistency.

Education

Master of Information Technology
RMIT University

2023 – 2024
Australia

Bachelor of Computer Science
Mingchuan University

2015- 2020
Taiwan

Other Skills and Licenses

Full working rights
Australia Drive license